

Convex Sets (1 pt)

Example sets

Sketch the following sets in $\mathbb{R}^2$

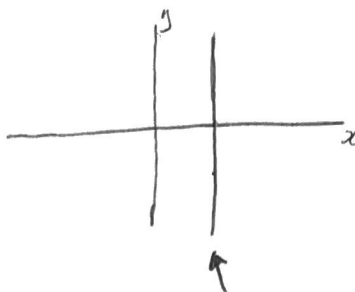
1. $\text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -0.5 \\ -0.5 \end{pmatrix} \right\}$

2. $\text{span} \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0.5 \\ -0.5 \end{pmatrix} \right\}$

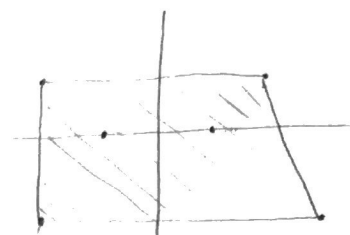
3. $\text{aff} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$

4. $\text{conv} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ -2 \end{pmatrix} \right\}$

3.)



4.)



Convexity

Let $C \subset \mathbb{R}^n$ be a convex set, with $x_1, \dots, x_k \in C$, and let $\theta_1, \dots, \theta_k \in \mathbb{R}$ satisfy $\theta_i \geq 0, \theta_1 + \dots + \theta_k = 1$. Show that $\theta_1 x_1 + \dots + \theta_k x_k \in C$. (The definition of convexity is that this holds for $k=2$; you must show it for arbitrary k .) Hint. Use induction on k .

Base. Let $k=2$: $\theta_1 x_1 + \theta_2 x_2$. $\theta_2 = 1 - \theta_1$ since $\theta_1 + \theta_2 = 1$. $\Rightarrow \theta_1 x_1 + (1 - \theta_1) x_2 \in C$ (by def. of convexity)

Induction. Assume $P_k = \theta_1 x_1 + \dots + \theta_k x_k \in C$. For $P_{k+1} = \theta_1 x_1 + \dots + \theta_k x_k + \theta_{k+1} x_{k+1}$, let $\theta_{k+1} = 0$, then $P_{k+1} = P_k \in C$. ■

Linear Equations

Show that the solution set of linear equations $\{x | Ax = b\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ is an affine set.

Let $S = \{x | Ax = b\}$ (i.e. $Ax_1 = b$ and $Ax_2 = b$)

Let $x_1, x_2 \in S$. Want to show that $\alpha_1 x_1 + \alpha_2 x_2 \in S$, i.e. $A(\alpha_1 x_1 + \alpha_2 x_2) = b$, with $\alpha_1 + \alpha_2 = 1$
 $A(\alpha_1 x_1 + (1 - \alpha_1) x_2) = \alpha_1 A x_1 + (1 - \alpha_1) A x_2 = \alpha_1 b + (1 - \alpha_1) b = b$ ■
 $\Rightarrow \alpha_2 = 1 - \alpha_1$

Linear Inequalities

1. Show that the solution set of linear inequalities $\{x | Ax \preceq b, Cx = d\}$ with $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k \times n}$ and $d \in \mathbb{R}^k$ is a convex set. Here $\preceq$ means componentwise less or equal.

Let $S = \{x | Ax \preceq b, Cx = d\}$ and $x_1, x_2 \in S$. Want to show $\alpha_1 x_1 + \alpha_2 x_2 \in S$, where $\alpha_1 + \alpha_2 = 1, \alpha_1, \alpha_2 \geq 0$
 $A(\alpha_1 x_1 + (1 - \alpha_1) x_2) = \alpha_1 A x_1 + (1 - \alpha_1) A x_2 \preceq \alpha_1 b + (1 - \alpha_1) b = b$ (since $\alpha_1, 1 - \alpha_1 \geq 0$)
 $C(\alpha_1 x_1 + (1 - \alpha_1) x_2) = \alpha_1 C x_1 + (1 - \alpha_1) C x_2 = \alpha_1 d + (1 - \alpha_1) d = d$ ■

2. Is it an affine set?

$$C = 0_{1 \times 1} \quad d = [0]$$

NO. Let $x_1 = [0]$, $x_2 = [1]$, $A = I_{1 \times 1}$, $b = [1]$. Clearly, $x_1, x_2 \in S$ (as defined in part 4.) (note: $x_1, x_2 \in \mathbb{R}^1$)

Let $x_1 = -1$, $x_2 = 2$: $A(-1[0] + 2[1]) = [2] > b$, hence $(x_1 x_1 + x_2 x_2) \notin S$ $\blacksquare$
($x_1 + d_2 = 1 \Rightarrow$ affine)

$\rightarrow$ either half of the space divided by a (hyper) plane.

Voronoi description of halfspace

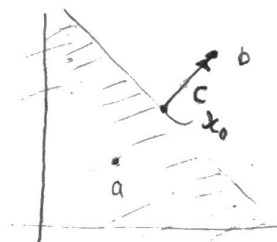
Let a and b be distinct points in $\mathbb{R}^n$. Show that the set of all points that are closer (in Euclidean norm) to a than b , i.e., $\{x \mid \|x - a\|^2 \leq \|x - b\|^2\}$, is a halfspace. Describe it explicitly as an inequality of the form $c^T x \leq d$. Draw a picture.

$$\vec{c} = \frac{1}{2}(b - a)$$

$$d = \vec{c}^T \vec{x}_0$$

$$\Rightarrow c^T x \leq d \Rightarrow c^T x - c^T x_0 \leq 0 \Rightarrow c^T (x - x_0) \leq 0 \Rightarrow \frac{1}{2}(b^T - a^T)(x - x_0) \leq 0$$

$$\text{where } x_0 = a + \frac{1}{2}(b - a)$$



Convex Illumination Problem (1 pt)

Show that the solution $p^* = (p_1^*, p_2^*, \dots, p_n^*)^T \in \mathbb{R}^n$ of the non-convex illumination problem from the lecture

$$\begin{aligned} &\text{minimize} && \max_{k=1 \dots m} |\log I_k - \log I_{des}| \\ &\text{subject to} && 0 \leq p_j \leq p_{max}, \quad j = 1 \dots n \end{aligned}$$